

Dr Alejandro Jiménez Rios

BSc, MSc, PgCert, PhD, FHEA

1. Contact Information

Academic Website: <https://researchportal.bath.ac.uk/en/persons/alejandro-jimenez-rios>
Personal Website: <https://jimnezra.github.io/>
Email: ajr225@bath.ac.uk
Phone Number: +44 (0) 1225 383719
Address: 6 East, University of Bath, Bath, BA2 7AY, United Kingdom
ORCID: <https://orcid.org/0000-0003-4470-255X>
Scopus Author ID: [57203578697](https://orcid.org/0000-0003-4470-255X)
Web of Science: [AAT-5218-2020](https://orcid.org/0000-0003-4470-255X)
Google Scholar: <https://scholar.google.es/citations?user=KMpFEg4AAAAJ&hl=en>
ResearchGate: <https://www.researchgate.net/profile/Alejandro-Jimenez-Rios-2>
LinkedIn: https://www.linkedin.com/in/jimnezra/?locale=en_US

2. Research Interests

My research interests include: the conservation of existing built cultural heritage assets through inspection, diagnosis, intervention design, and monitoring; the study of vernacular architecture, regenerative design practices, and nature-based solutions; the assessment of mechanical properties and structural behaviour of masonry, bamboo, and earthen structures through the application of minor and non-destructive experimental testing, specialized constitutive models, and advanced non-linear numerical simulations; the enhancement of structural robustness and resilience; as well as the development and implementation of Industry 5.0 principles and enabling technologies in the architecture, construction, engineering, management, operation, and conservation industry such as artificial intelligence, the internet of things, structural health monitoring, asset information modelling, and digital twins. With my research, I aim at contributing to the achievement of United Nations Sustainable Development Goals 8, 9, 11, and 13.

3. Teaching Philosophy

My teaching philosophy is centred around creating an active, inclusive, and safe learning environment that fosters the creation of deep learning knowledge among my students. My goal is to develop on my students the skills and knowledge necessary for success in the field and to empower them with learning tools that will allow them to take ownership of their own learning and further professional development. I conduct myself with integrity following strict codes of conduct and strive to adopt diversity, equity, and inclusion principles by respecting other's identity, nationality, socioeconomic level, gender, sexual orientation, disabilities, and age. I am interested in implementing a Scholarship of Teaching and Learning approach on my teaching practice. With my teaching, I aim at contributing to the achievement of United Nations Sustainable Development Goal 4. I am an awarded Fellow of Advance HE (FHEA) in alignment with the UK Professional Standards Framework (PSF 2023).

4. Current Positions

05/08/2024 - Present: Lecturer in Structural Engineering.
01/08/2025 – Present: [History, Conservation, and Management Cluster](#) Co-Leader.
Department: [Department of Architecture & Civil Engineering](#).
Research Centres: [Centre for Regenerative Design & Engineering for a Net Positive World \(RENEW\)](#).
[Centre for Digital, Manufacturing & Design \(The Foundry\)](#).
[Centre for Climate Adaptation & Environment Research \(CAER\)](#).
[Centre of Excellence in Engineering & Design Education \(CEEDE\)](#).
Faculty: [Faculty of Engineering & Design](#).
University: [University of Bath](#).
Country: United Kingdom.

5. Education and Qualifications

- 2026:** Fellow of Advance HE (FHEA).
Organisation: Advance HE, United Kingdom.
Description: Fellowship demonstrates a personal and institutional commitment to professionalism in learning and teaching in higher education.
- 01/09/2016 - 09/04/2020:** PhD in Civil, Structural and Environmental Engineering.
University: Trinity College Dublin, Ireland.
Thesis: Conservation of earthen vernacular architecture in Ireland. Study of the mechanical properties and the structural behaviour of cob.
Funding: Internal University funding Studentship Award.
- 01/09/2017 - 31/08/2018:** Postgraduate Certificate in Statistics.
University: Trinity College Dublin, Ireland.
Description: Base module, introduction to regression and design of experiments.
Funding: Internal University funding Studentship Award.
- 01/09/2015 - 31/08/2016:** Erasmus Mundus Advanced Master in Structural Analysis of Monuments and Historical Constructions (SAHC).
Universities: Czech Technical University, Czechia and Polytechnic University of Catalonia, Spain.
Thesis: Simulation of masonry structural behaviour using the Discrete Element Method.
Funding: Erasmus+ Scholarship.
- 01/09/2009 - 31/08/2014:** Bachelor's Degree in Civil Engineering, Structural Engineering Specialization Path.
University: University of Guadalajara, Mexico.
Thesis: Comparative analysis of plane frames' structural behaviour considering the effect of end length offsets.
- 01/07/2012 - 30/06/2013:** International Academic Exchange.
University: National Institute of Applied Sciences (INSA) Lyon, France.
Funding: Mexfitec Scholarship.

6. Research Experience

- 01/04/2026 - 31/03/2027:** Principal Investigator (User Group Leader).
Institution: University of Bath, United Kingdom.
Project: High-Resolution Infrared Thermography for Structural Assessment of Historic Fibrous Plaster Ceilings at the Royal Albert Hall.
Funding: 1st Call for Access to E-RIHS Services.
- 01/10/2025 - 31/12/2026:** CoSeC Fellow.
Institution: University of Bath, United Kingdom.
Project: [Safe Digital Twins for the Built Environment \(SafeDT\)](#).
Funding: 2025 Computational Science Centre for Research Communities (CoSeC) Fellowship Programme.
- 01/06/2026 - 31/07/2026:** Principal Investigator.
Institution: University of Bath, United Kingdom.
Project: [Regenerative Retrofitting Via Ecologically Active Soil Structures II \(REECO-SOIL II\)](#).
Funding: 2026 RENEW Summer Bursary.
- 01/06/2026 - 31/07/2026:** Co-Investigator.
Institution: University of Bath, United Kingdom.
Project: Place-based approaches to regenerative education: using walking guides to reveal the story of a place.
Funding: 2026 RENEW Summer Bursary.

- 01/06/2026 - 31/07/2026:** Co-Investigator.
Institution: University of Bath, United Kingdom.
Project: CrackVision: An AI-Driven Approach to Monitoring Wall Condition and Damage Severity.
Funding: 2026 The Foundry Summer Bursary.
- 01/06/2025 - 31/07/2025:** Principal Investigator.
Institution: University of Bath, United Kingdom.
Project: [Regenerative Retrofitting Via Ecologically Active Soil Structures \(REECO-SOIL\).](#)
Funding: 2025 RENEW Summer Bursary.
- 01/06/2025 - 31/07/2025:** Co-Investigator.
Institution: University of Bath, United Kingdom.
Project: [Life Cycle Assessment and Optimisation of Waste Stone Structures](#)
Funding: 2025 ISCC Summer Bursary.
- 01/05/2024 - 04/08/2024:** Visiting Researcher.
Institution: Delft University of Technology, Netherlands.
Description: Hosted at the Faculty of Civil Engineering and Geosciences, I worked on the Secondment of the DTADD project.
Funding: Marie Skłodowska-Curie Actions (MSCA) grant agreement No 101066739.
- 01/02/2024 - 30/04/2024:** ICCROM Fellow.
Institution: International Centre for the Study of the Preservation and Restoration of Cultural Property (ICCROM).
Description: During this fellowship I conducted a systematic literature review and an online survey to explore how the architecture, engineering, construction, management, operation, and conservation (AECMO&C) industry could adapt and adopt novel Industry 5.0 principles and enabling technologies, ultimately enhancing built cultural heritage conservation practices.
Funding: Research Stay Abroad Scholarship, Research Council of Norway.
- 01/11/2022 - 31/01/2024:** MSCA Postdoctoral Research Fellow.
University: Oslo Metropolitan University, Norway.
Description: My MSCA fellowship had two specific objectives. The first one is to build digital twin (DT) models of heritage/conventional bridges to assess and identify the highest performing anomaly detection algorithm (ADA) for damage and/or significant decay detection. Its second objective is to develop an ADA-informed open-source decision-making tool based on a reliability-based bridge management approach, to assess the need for bridge intervention while explicitly considering their cultural heritage value.
Funding: Marie Skłodowska-Curie Actions (MSCA) grant agreement No 101066739.
Website: <https://alejandroslomet.github.io/>
- 01/11/2020 - 31/10/2022:** Postdoctoral Researcher.
University: Sapienza University of Rome, Italy.
Description: The research project in which I participated focused on the study of the structural behaviour of masonry with the use of a non-standard limit analysis method implemented through the ALMA 2.0 program and on this software extension.
Funding: Internal University funding.
- Summer 2018:** Visiting Researcher.
University: University of Cantabria, Spain.
Description: International research collaboration to design the experimental campaign on the use of minor-destructive tests to determine the mechanical properties of cob.
Funding: Internal University funding Studentship Award.

7. Received Funding/Fellowships/Awards

- **2026 RENEW Summer Bursary:** *Centre for Regenerative Design & Engineering (RENEW)*. £3000 awarded to hire a research assistant for 8 weeks and pay for direct expenses during the summer 2026.
- **2026 The Foundry Summer Bursary (as CoPI):** *Centre for Digital, Manufacturing & Design (The Foundry)*. £3000 awarded to hire a research assistant for 8 weeks and pay for direct expenses during the summer 2026.
- **2026 RENEW Summer Bursary (as CoPI):** *Centre for Regenerative Design & Engineering (RENEW)*. £3000 awarded to hire a research assistant for 8 weeks and pay for direct expenses during the summer 2026.
- **2026 Impact Fund.** *University of Bath*. £3000 to attend the international academic conference Terra 2026, taking place at Al Ain, UAE; present previously completed and ethically approved research; engage in networking and professional knowledge-exchange activities; and hold exploratory meetings with academic and professional partners (e.g., ICOMOS ISCEAH, ICCROM Sharjah, University of Sharjah).
- **2025 CoSeC Fellowship.** *Computational Science Centre for Research Communities (CoSeC)*. £3000 to run an idea factory event to identify important cross-cutting topics, explore new ideas and set roadmaps for the project “Safe Digital Twins for the Built Environment (SafeDT)”.
- **2025 Amazonia BR/UK Workshop Grant:** *British Council*. Around £3000 awarded to cover the costs related to the participation to the workshop.
- **2025 ISCC Summer Bursary (as CoPI):** *Institute of Sustainability and Climate Change (ISCC)*. £3000 awarded to hire a research assistant for 8 weeks and pay for direct expenses during the summer 2025.
- **2025 RENEW Summer Bursary:** *Centre for Regenerative Design & Engineering (RENEW)*. £3000 awarded to hire a research assistant for 8 weeks and pay for direct expenses during the summer 2025.
- **2024 MCAA Micro Grant:** *Marie Curie Alumni Association*. 820 € received to finance my participation in the EuroScience Open Forum (ESOF) 2024 at Katowice, Poland.
- **2024 Research Stay Abroad Scholarship:** *Research Council of Norway*. Around 8000 € received to finance my stay during my ICCROM Fellowship in Rome, Italy, for the period from 01 March to 30 April 2024.
- **2024 ICCROM Visiting Fellowship:** *International Centre for the Study of the Preservation and Restoration of Cultural Property (ICCROM)*. Fellowship awarded to conduct a research proposal on “Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice”, for the period from 01 February to 30 April 2024.
- **2023 Project Establishment Support:** *Research Council of Norway*. Around 5000 € received after the successful completion and submission of a European Research Council Starting Grant (ERC StG).
- **2023 Erasmus + Staff Mobility Grant:** *Oslo Metropolitan University*. Around 1200 € which covered course fees, transportation, accommodation, and food expenses to participate in the Academic Leadership course, delivered by the European Academy of Innovation.
- **2022-2024 Marie Skłodowska-Curie Actions Individual Fellowship:** *Oslo Metropolitan University/Delft University of Technology*. Around 227,000 € which cover living allowance, mobility allowance, family allowance, research, training, and networking contributions as well as management and indirect contributions.
- **2020-2021 Internal University funding:** *Sapienza University of Rome*. Around 50,000 € which covered health insurance, travel expenses and a monthly stipend of 1600 €.
- **2016-2019 Internal University funding Studentship Award:** *Trinity College Dublin*. Around 100,000 € which covered tuition fees, research expenses and a monthly stipend of 1 200 €.
- **2015-2016 Erasmus+ Scholarship:** *Czech Technical University/ Polytechnic University of Catalonia*. Around 25,000 € which covered tuition fees, health insurance and a monthly stipend of 1 000 €.
- **2012-2013 Mexfitec Scholarship:** *University of Guadalajara/ National Institute of Applied Sciences Lyon*. Around 20,000 €, which covered tuition fees, return flight Mexico-France, two-months French course, health insurance and a monthly stipend of 800 €.

8. Tracked Record

As of 06/06/2026 I have published a total of 25 peer-reviewed journal papers, 20 peer-reviewed conference papers, 1 peer-reviewed book chapter, 4 conference posters or presentations, 3 dissertations, 14 databases and several other open science related and non-peer-reviewed publications.

Scopus: h-index:10, citations:378, documents: 43.
Web of Science: h-index:9, citations:258, documents: 26.
ResearchGate: h-index: 11 citations: 554, research elements: 70.
Google Scholar: h-index:12, citations:584, documents: 87.

9. Publications

Peer-reviewed scientific journal papers:

1. Shaban, N., Arango, E., **Jiménez Rios, A.**, Rincon, L. F., Shang, Y., Habeeb, B., Martín-Pérez, B., Ferreira, T. M., Mendoza-Lugo, M. A., Ramírez Eudave, R., Bastidas-Arteaga, E. Sousa, H. S., & Nogal, M. (2026) Perspectives on the resilience of existing bridges to short- and long-term impacts of climate change. Structure and Infrastructure Engineering. *The paper synthesises interdisciplinary research on climate change challenges and highlights actionable adaptation strategies to enhance resilience at both asset and network levels.* [DOI: 10.1080/15732479.2026.2651414](https://doi.org/10.1080/15732479.2026.2651414).
2. Cin Man, A. L., & **Jiménez Rios, A.** (2026) Tourist Taxes for Public Realm Conservation in Bath: A Stakeholder Analysis. The Historic Environment: Policy & Practice. *From this work, it was found the relationship between the Bath local authority and lobbying parties, businesses, tourists, and residents already has a strong basis for continued collaboration.* [DOI: 10.1080/17567505.2026.2656047](https://doi.org/10.1080/17567505.2026.2656047).
3. Mashal, M., & **Jiménez Rios, A.** (2026) A Benchmark Model for Earthquake-Resistant Earthen Houses in Rural Afghanistan. Buildings. *This study synthesises research findings and global earthen building codes and guidelines to develop a practical, context-appropriate benchmark house model for Afghanistan.* [DOI: 10.3390/buildings16010011](https://doi.org/10.3390/buildings16010011).
4. Thongchom, C., Ghamari, A., **Jiménez Rios, A.**, & Karimi, I. (2025) Assessment of the performance of a novel T-Stub-enhanced ConXL moment connection under fire. Emergency Management Science and Technology. *This study proposes an innovative enhancement to the ConXL connection by incorporating a T-stub for application with unfilled box columns. The enhanced connection is analysed through parametric and numerical investigations, with a particular focus on its post-fire behaviour.* [DOI: 10.48130/emst-0025-0021](https://doi.org/10.48130/emst-0025-0021).
5. **Jiménez Rios, A.**, Plevris, V., & Nogal, M. (2025) Towards Industry 5.0: a stakeholder analysis to understand the human role in the adoption of a heritage bridge human-centric digital twin framework. Structure and Infrastructure Engineering. *In this article, the methodology and the findings obtained after conducting a stakeholder analysis to understand how different actors could work together towards the adoption of Industry 5.0 principles and enabling technologies are presented.* [DOI: 10.1080/15732479.2025.2490126](https://doi.org/10.1080/15732479.2025.2490126).
6. **Jiménez Rios, A.**, Plevris, V., Nogal, M., & Admiraal, W. (2025) Scholarship of Teaching and Learning in Civil and Structural Engineering: A Systematic Literature Review. Journal of Civil Engineering Education. *This systematic literature review addresses the following questions: (1) In what capacity do educators within the field of civil and structural engineering (CaSE) engage with SoTL?, and (2) What are the benefits of implementing a SoTL for CaSE educators?* [DOI: 10.1061/JCEECD.EIENG-210](https://doi.org/10.1061/JCEECD.EIENG-210).
7. **Jiménez Rios, A.**, Nela, B., Pingaro, M., Reccia, E., & Trovalusci, P. (2025) Parametric analysis of masonry arches following a limit analysis approach: influence of joint friction, pier texture and arch shallowness. Mathematics and Mechanics of Solids. *In this paper, parameters such as pier texture, joint friction angle, and arch shallowness, namely shallow, semi-circular, and pointed arches, were investigated under three scenarios: horizontal, concentrated vertical live load applied at mid-span and quarter-span.* [DOI: 10.1177/10812865231175385](https://doi.org/10.1177/10812865231175385).
8. **Jiménez Rios, A.**, Nogal, M., Plevris, M., Ramirez, R., & L. Petrou, M. (2024) Towards Enhanced Built Cultural Heritage Conservation Practices: Perceptions on Industry 5.0 Principles and Enabling Technologies. The Historic Environment: Policy & Practice. *This study aims to gain a deeper understanding of conservation*

professionals' perceptions regarding the adoption of Industry 5.0 principles and enabling technologies, as well as the perceived barriers and the skills needed to address them. [DOI: 10.1080/17567505.2024.2429167](https://doi.org/10.1080/17567505.2024.2429167).

9. **Jiménez Rios, A.,** L. Petrou, M., Ramirez, R., Plevris, V., & Nogal, M. (2024) Industry 5.0, Towards an Enhanced Built Cultural Heritage Conservation Practice. *Journal of Building Engineering*. *The purpose of this article is to shed light on how the architecture, engineering, construction, management, operation, and conservation industry can adapt and better prepare to embrace novel Industry 5.0 principles and enabling technologies, ultimately resulting in enhanced conservation practices for the built cultural heritage environment.* [DOI: 10.1016/j.jobe.2024.110542](https://doi.org/10.1016/j.jobe.2024.110542).
10. **Jiménez Rios, A.,** Ben Seghier, M. E. A., Plevris, V., & Dai, J. (2024) Explainable ensemble learning framework for estimating corrosion rate in suspension bridge main cables. *Results in Engineering*. *In this study, a robust framework is proposed to predict the annual corrosion rate in main cables of suspension bridges, while investigating the impact of the surrounding environmental factors on this process.* [DOI: 10.1016/j.rineng.2024.102723](https://doi.org/10.1016/j.rineng.2024.102723).
11. Arce, A., **Jiménez Rios, A.,** Tomic, I., & Biggs, D. (2024) Structural Analysis of the Sympathetic Restoration and Conservation of the Gopinath Temple, Kathmandu, Nepal. *Heritage*. *In this paper, the case study of the earthquake-damaged Gopinath temple is discussed. First, an initial visual inspection phase and the following diagnosis of the structure are discussed. A sympathetic intervention proposal capable of increasing the temple's safety level, and based on the addition of timber plates, has resulted in substantial improvements in the lateral behavior of the structure.* [DOI: 10.3390/heritage7060151](https://doi.org/10.3390/heritage7060151).
12. **Jiménez Rios, A.,** Petrou, M. L., Ramirez, R., Plevris, V., & Nogal, M. (2024) Industry 5.0 concepts and enabling technologies, towards an enhanced conservation practice: systematic literature review protocol. *Open Research Europe*. *This paper outlines the protocol adopted to conduct a systematic literature review with the aim of exploring how the Architecture, Engineering, Construction, Management, Operation, and Conservation (AECMO&C) industry can adapt and be better prepared to embrace novel Industry 5.0 principles and enabling technologies, ultimately resulting in enhanced conservation practices for the built cultural heritage environment.* [DOI: 10.12688/openreseurope.17505.1](https://doi.org/10.12688/openreseurope.17505.1).
13. Nela, B., **Jiménez Rios, A.,** Pingaro, M., Reccia, E., & Trovalusci, P. (2023) Masonry arches simulations using cohesion parameter as code enrichment for limit analysis approach. *International Journal of Masonry Research and Innovation*. *In this paper, the cohesion incorporation is calibrated for a variety of in-plane applications, accounting for the joints' indirect tensile strength, that can simulate the strengthening measures.* [DOI: 10.1504/IJMRI.2024.135236](https://doi.org/10.1504/IJMRI.2024.135236).
14. **Jiménez Rios, A.,** Georgioudakis, M. S., Song, Y., & Ruiz-Capel, S. (2023) Editorial: Methods and Applications in Computational Methods in Structural Engineering. *Frontiers in Built Environment*. *Editorial of the "Computational Methods in Structural Engineering" Research Topic in Frontiers in Built Environment journal which highlights the transformative power of computational methods, showcasing their potential to revolutionize the way we design, analyze, and assess the structures.* [DOI: 10.3389/fbuil.2023.1339541](https://doi.org/10.3389/fbuil.2023.1339541).
15. Plevris, V., Papazafeiropoulos, G., & **Jiménez Rios, A.,** (2023) Chatbots Put to the Test in Math and Logic Problems: A Comparison and Assessment of ChatGPT-3.5, ChatGPT-4, and Google Bard. *AI. Our study unveils the secrets behind three digital wizards, ChatGPT-3.5, ChatGPT-4, and Google Bard, as they engage in a thrilling showdown of mathematical and logical prowess.* [DOI: 10.3390/ai4040048](https://doi.org/10.3390/ai4040048).
16. **Jiménez Rios, A.,** (2023) Learning from the Past: Parametric Analysis of Cob Walls. *Applied Sciences*. *In this paper, a series of response surfaces and parametric equations with which it is possible to compute safety factors and collapse multipliers of existing cob vernacular walls (within the range of values studied) are provided.* [DOI: 10.3390/app13159045](https://doi.org/10.3390/app13159045).
17. **Jiménez Rios, A.,** Ruiz-Capel, S., Plevris, V., & Nogal, M. (2023) Computational Methods Applied to Earthen Historical Structures. *Frontiers in Built Environment*. *This paper compiles, summarizes, and highlights the latest developments and implementations of computational methods for the study of earthen historical structures. Future trends, opportunities, and challenges are discussed.* [DOI: 10.3389/fbuil.2023.1219108](https://doi.org/10.3389/fbuil.2023.1219108).
18. **Jiménez Rios, A.,** Plevris, V., & Nogal, M. (2023) Bridge management through digital twin-based anomaly detection systems: A systematic review. *Frontiers in Built Environment*. *This paper, collects and synthesizes state-of-the-art knowledge and information about how bridge information modelling, finite elements, and bridge health monitoring are combined and used in the creation of digital twins (DT) of bridges, and how*

these models could generate damage scenarios to be used by anomaly detection algorithms for damage detection on bridges, especially in those bridges with cultural heritage. [DOI: 10.3389/fbuil.2023.1176621](https://doi.org/10.3389/fbuil.2023.1176621).

19. Nela, B., **Jiménez Rios, A.**, Pingaro, M., Reccia, E., & Trovalusci, P. (2022) Limit analysis of locally reinforced masonry arches. Engineering Structures. *In this paper, the influence of partial reinforcement on the structural response of masonry arches is studied.* [DOI: 10.1016/j.engstruct.2022.114921](https://doi.org/10.1016/j.engstruct.2022.114921).
20. **Jiménez Rios, A.**, Nela, B., Pingaro, M., Reccia, E., & Trovalusci, P. (2022) Rotation and sliding collapse mechanisms for in plane masonry pointed arches: statistical parametric assessment. Engineering Structures. *In this paper, the effects that different geometrical (slenderness and sharpness) and mechanical (friction and cohesion) parameters have on the in-plane structural response of masonry pointed arches are investigated. Two load scenarios are studied: vertical self-weight plus a proportional horizontal live load and vertical self-weight plus a vertical concentrated live load applied to individual voussoir.* [DOI: 10.1016/j.engstruct.2022.114338](https://doi.org/10.1016/j.engstruct.2022.114338).
21. **Jiménez Rios, A.**, Pingaro, M., Reccia, E., & Trovalusci, P. (2022) Statistical assessment of in-plane masonry panels using limit analysis with sliding mechanism. ASCE Journal of Engineering Mechanics. *In this paper the effects that different geometrical (panel ratio, block ratio, bond type) and mechanical (friction ratio) parameters have on the in-plane structural response of brick masonry panels are investigated.* [DOI: 10.1061/\(ASCE\)EM.1943-7889.0002061](https://doi.org/10.1061/(ASCE)EM.1943-7889.0002061).
22. **Jiménez Rios, A.**, & O'Dwyer, D. (2020) Experimental validation of the application of the flat jack in cob walls. Construction and building materials. *The purpose of the experimental campaign presented in this paper was to assess the feasibility of the technique to measure the average compressive levels of stress in existing cob walls and determine their mechanical properties.* [DOI: 10.1016/j.conbuildmat.2020.119148](https://doi.org/10.1016/j.conbuildmat.2020.119148).
23. **Jiménez Rios, A.**, & O'Dwyer, D. (2020) Data collected from the experimental validation for the application of flat jack tests in cob walls. Data in brief. *It contains data collected during the consolidation process of six cob wallettes and data collected after single and double flat jack test were applied for the first time to a set of cob wallettes at the laboratory facilities of the Department of Civil, Structural and Environmental Engineering at Trinity College Dublin.* [DOI: 10.1016/j.dib.2020.105764](https://doi.org/10.1016/j.dib.2020.105764).
24. **Jiménez Rios, A.**, & O'Dwyer, D. (2020). Adaptations of the flat jack test for its application in cob walls. MethodsX. *Two standards have been developed for the application of the technique in masonry walls and this paper presents the adaptation of those methods for its application in cob walls.* [DOI: 10.1016/j.mex.2020.101003](https://doi.org/10.1016/j.mex.2020.101003).
25. **Jiménez Rios, A.**, & O'Dwyer, D. (2020) Numerical modelling of cob's non-linear monotonic structural behaviour. International Journal of Computational Methods. *This paper presents the results obtained of the simulation of cob's nonlinear monotonic behaviour using two well-known finite element commercial packages.* [DOI: 10.1142/S0219876219500464](https://doi.org/10.1142/S0219876219500464).

Peer-reviewed conference papers published in conference proceedings:

1. **Jiménez Rios, A.**, Calabria-Holley, J., & Castilla Pascual, F. J. (2026) Regenerative Retrofitting of Built Environment Assets via Ecologically Active Soils. Presented at the 1st RILEM Youth Symposium on Building Materials and Construction (RYS2025). Online. *This paper presents the rationale and the objectives of the Regenerative Retrofitting Via Ecologically Active Soil Structures (Reeco-Soil) project, which aims to establish a comprehensive understanding of the state-of-the-art and regenerative potential of retrofitting built environment assets using ecologically active soil structures.* [DOI: 10.1007/978-3-032-14652-6_40](https://doi.org/10.1007/978-3-032-14652-6_40).
2. Losini, A. M., Walter Bruno, A., Posani, M., Armistead, S. J., Babafemi, A. J., Faria, P., Guihéneuf, S., Ioannou, I., **Jiménez Rios, A.**, Kamath, M., Mwebaza, D., Panagiotou, R., Ranesi, A., Theodoridou, M., Vaculik, S., Vucetic, S., Habert, G., Perlot, C., & Armada Brás, A. M. (2026) Recent Advancements on the Effect of Bio-Stabilisation on the Mechanical and Hygrothermal Properties of Earthen Building Materials. Presented at the 1st RILEM Youth Symposium on Building Materials and Construction (RYS2025). Online. *This paper provides an overview of existing experimental methods for measuring the mechanical, hygroscopic and thermal properties of bio-stabilised earthen composites and products, along with key findings from the literature.* [DOI: 10.1007/978-3-032-14652-6_38](https://doi.org/10.1007/978-3-032-14652-6_38).
3. Posani, M., Habert, G., Fogue Djombou, Y. I., Du, Y., Vereecken, E., Esteve-Bourrel, P., Pinchard, L., Losini, A. E., Barreto Arez Coelho, G., Ben-Alon, L., Beatrice Carcassi, O., Dormohmadi, M., **Jiménez**

- Rios, A.,** Kamath, M., Libralato, M., Maia, J., Makoundou, C., Malet-Damour, B., Ranesi, A., Perlot, C., Walter Bruno, A., Vučetić, S., & Armada Brás, A. M. (2026) Bio-Stabilised Earthen Materials: A Perspective on the Potential Contribution to Climate Change Adaptation and Mitigation. Presented at the 1st RILEM Youth Symposium on Building Materials and Construction (RYS2025). Online. *This paper introduces the work of the RILEM TC BEC and presents findings from its literature review, focusing on the potential of bio-stabilised earth for climate change adaptation and mitigation.* [DOI: 10.1007/978-3-032-14652-6_32](https://doi.org/10.1007/978-3-032-14652-6_32).
4. **Jiménez Rios, A.,** Ramirez, R., L. Petrou, M., Plevris, V., & Nogal, M. (2026) Advancing Built Cultural Heritage Conservation: Integration of Industry 5.0 Principles and Enabling Technologies. Presented at the 14th International Conference on Structural Analysis of Historical Constructions (SAHC 2025), Lausanne, Switzerland. *This article explores the implications of Industry 5.0 principles and enabling technologies within the Architecture, Engineering, Construction, Management, Operation, and Conservation (AECMO&C) industry, with a particular focus on the conservation of built cultural heritage environments.* [DOI: 10.1007/978-3-032-16767-5_58](https://doi.org/10.1007/978-3-032-16767-5_58).
 5. Plevris, V., & **Jiménez Rios, A.** (2025) Navigating Uncertainty in Image-Based Structural Health Monitoring: The Role of False Positives, False Negatives, and Base Rate Bias. Presented at the 6th International Conference on Uncertainty Quantification in Computational Sciences and Engineering (UNCECOMP). *Using numerical simulations, the paper illustrates how base rate bias reduces precision in low-damage environments and explores the broader implications for system design, operational risk, and decision-making.* [DOI: 10.7712/120225.12384.21569](https://doi.org/10.7712/120225.12384.21569).
 6. Ramirez, R., **Jiménez Rios, A.,** Ghiassi, B., & Lourenco, P. B. (2025) Modeling Approaches for Moisture Transport in Brick Masonry Walls. Presented at the International Conference on Moisture in Buildings (ICMB), Guimarães, Portugal. *This study highlights the impact of anisotropic transport behavior and emphasize the need for further refinement of simplified modelling strategies, particularly through the incorporation of interface hydraulic resistances.* [DOI: 10.1007/978-3-032-09054-6_50](https://doi.org/10.1007/978-3-032-09054-6_50).
 7. Plevris, V., **Jiménez Rios, A.,** & Ebead, U. A. (2025) Exploring the Predictive Performance of Simple Regression Models and ANN in 2D Truss Analysis. Presented at the 1st International Conference on Net-Zero Built Environment, Oslo, Norway. *This research investigates the performance of various regression models in predicting critical structural parameters within a plane truss model.* [DOI: 10.1007/978-3-031-69626-8_123](https://doi.org/10.1007/978-3-031-69626-8_123).
 8. Plevris, V., Almutairi, A., & **Jiménez Rios, A.** (2025) Advancing Sustainability Through Structural Optimization: Innovations in Material Efficiency and Environmental Impact Reduction. Presented at the 1st International Conference on Net-Zero Built Environment, Oslo, Norway. *This work explores how the strategic use of materials—not merely reducing their quantity but optimizing their application—can lead to structures that uphold or enhance quality, safety, and functionality, while significantly diminishing the environmental footprint of construction activities.* [DOI: 10.1007/978-3-031-69626-8_134](https://doi.org/10.1007/978-3-031-69626-8_134).
 9. Ramirez, R., **Jiménez Rios, A.,** Ghiassi, B., & Lourenco, P. B. (2024) Application of Mechanical Modelling Strategies to the Hygrothermal Analysis of Masonry Walls. Presented at the 9th European Congress on Computational Methods in Applied Sciences and Engineering ECCOMAS Congress 2024, Lisbon, Portugal. *This study investigates the hygrothermal transport mechanisms in brick masonry, focusing on the applicability of different modelling approaches, namely detailed micro-modelling, continuous micro-modelling, discrete micro-modelling, and macro-modelling.* [DOI: 10.23967/eccomas.2024.155](https://doi.org/10.23967/eccomas.2024.155).
 10. **Jiménez Rios, A.,** Demirlioglu, K., Plevris, V., & Nogal, M. (2024) Towards the Creation of Synthetic Bridge Digital Twins What-If Scenarios: Study Case and Calibration. Presented at the 9th European Congress on Computational Methods in Applied Sciences and Engineering ECCOMAS Congress 2024, Lisbon, Portugal. *In this paper we present the S101 Bridge, which has been selected as first case study for the generation of the proposed synthetic database.* [DOI: 10.23967/eccomas.2024.091](https://doi.org/10.23967/eccomas.2024.091).
 11. **Jiménez Rios, A.** (2024) First findings on the mechanical analysis of cob using a Discontinuity Layout Optimization (DLO) approach. Presented at the 2nd RILEM International Conference on Earthen Construction, Edinburgh, United Kingdom. *In this paper, an alternative simulation tool known as Discontinuity Layout Optimization (DLO) has been explored to try to simulate cob's mechanical behaviour.* [DOI: 10.1007/978-3-031-62690-6_24](https://doi.org/10.1007/978-3-031-62690-6_24).
 12. **Jiménez Rios, A.,** Plevris, V., & Nogal, M. (2024) Human role in existing bridge digital twin frameworks, towards Industry 5.0. Presented at the 12th International Conference on Bridge Maintenance, Safety and Management, Copenhagen, Denmark. *In this paper we explore the increasingly important role that humans*

will have within existing bridge digital twin frameworks and how the human-centrism principle could be better implemented within such context. [DOI: 10.1201/9781003483755](https://doi.org/10.1201/9781003483755).

13. Ben Seghier, M. E. A., **Jiménez Rios, A.**, Plevris, V., & Dai, J. (2024) Advanced framework for degradation modelling of operating structures. Presented at the 12th International Conference on Bridge Maintenance, Safety and Management, Copenhagen, Denmark. *This research endeavours to predict the extent of corrosion in the primary cables of bridges through the application of advanced methodologies based on machine learning techniques.* [DOI: 10.1201/9781003483755](https://doi.org/10.1201/9781003483755).
14. **Jiménez Rios, A.**, Plevris, V., & Nogal, M. (2023) Uncertainties in the synthetic data generation for the creation of bridge digital twins. Presented at the 5th International Conference on Uncertainty Quantification in Computational Science and Engineering, Athens, Greece. *This paper explores the uncertainties involved in the synthetic data generation methodologies and tools, which could be used as a faster and cheaper alternative to real monitoring, for the creation and development of DT prototypes of bridges for both industry and research-oriented purposes.* [DOI: 10.7712/120223.10323.20020](https://doi.org/10.7712/120223.10323.20020).
15. **Jiménez Rios, A.**, Plevris, V., & Nogal, M. (2023) Synthetic data generation for the creation of bridge digital twins what-if scenarios. Presented at the 9th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Athens, Greece. *This paper aims at exploring the currently available synthetic data generation methodologies and tools, which could be used as a faster and cheaper alternative to real monitoring, for the creation and development of DT prototypes of bridges for both industry and research-oriented purposes.* [DOI: 10.7712/120123.10760.21262](https://doi.org/10.7712/120123.10760.21262).
16. Nela, B., Pingaro, M., **Jiménez Rios, A.**, Reccia, E., & Trovalusci, P. (2022) Masonry simulations using cohesion parameter as code enrichment for a non-standard limit analysis approach. Presented at the 7th Mechanics of Masonry Structures Strengthened with Composite Materials – Online Conference. *The cohesion incorporation is calibrated for a variety of in-plane applications, accounting for the joints' indirect tensile strength that can simulate the strengthening measures.* [DOI: 10.5281/zenodo.7496239](https://doi.org/10.5281/zenodo.7496239).
17. **Jiménez Rios, A.**, Grimes, M. & O'Dwyer, D. W. (2021) Experimental campaign on the use of the flat jack test in cob walls. Presented at the 12th International Conference on Structural Analysis of Historical Constructions – Online Conference. *This paper presents the first experimental campaign performed to validate the use of this Minor Destructive Technique (MDT) to determine the levels of stress in cob walls and their mechanical properties.* [DOI: 10.23967/sahc.2021.099](https://doi.org/10.23967/sahc.2021.099).
18. **Jiménez Rios, A.**, & O'Dwyer, D. W. (2019) External post-tensioning system for the strengthening of historical stone masonry bridges. Presented at the 11th International Conference on Structural Analysis of Historical Constructions, Cuzco, Peru. *The objective of this paper is to present the evaluation of the feasibility of the use of an external post-tensioning arrangement to strengthen historical stone masonry bridges. The arrangement proposed is an alternative to those already existent and developed by other authors.* [DOI: 10.1007/978-3-319-99441-3_168](https://doi.org/10.1007/978-3-319-99441-3_168).
19. **Jiménez Rios, A.**, & O'Dwyer, D. (2018) Earthen buildings in Ireland. Presented at the 6th International Congress on Construction History, Brussels, Belgium. Shortlisted within the top 10 best papers presented by PhD Students. *The objective of this paper is to present a summarized and concise picture of the present situation of the remaining earthen buildings in the country, identify their main characteristics, the values that make such buildings important, evaluate their vulnerability as vernacular architectural style and therefore, understand and better approach any future intervention on such kinds of structures.* [DOI: 10.5281/zenodo.7494070](https://doi.org/10.5281/zenodo.7494070).
20. **Jiménez Rios, A.**, & O'Dwyer, D. (2018) FEM non-linear modelling of cob using ANSYS. Presented at the 9th International Conference on Computational Methods, Rome, Italy. *The aim of this paper is to evaluate the available material models' suitability for the modelling of cob's structural behaviour in one of the most used FEM software in the market.* [DOI: 10.5281/zenodo.7494038](https://doi.org/10.5281/zenodo.7494038).

Pee-reviewed book chapters:

1. **Jiménez Rios, A.** (2025) Cob Vernacular Architecture Conservation and Its Contribution Towards a Sustainable Built Environment. In book: Sustainable Materials for the Built Environment. Editor: Dorian Hanaor. Springer. *In this chapter, the importance of cob vernacular heritage conservation and how this can contribute towards a sustainable built environment with an emphasis on the sustainable properties of this material is discussed.* [DOI: 10.1007/978-3-031-97818-0_6](https://doi.org/10.1007/978-3-031-97818-0_6).

Dissertations:

1. **Jiménez Rios, A.** (2020) Conservation of earthen vernacular architecture in Ireland. Study of the mechanical properties and the structural behaviour of cob. PhD thesis. Trinity College Dublin, Ireland. *The thesis presents a summarized and concise picture of the present situation of the remaining earthen buildings in Ireland, identifies their main characteristics, the values that make such buildings important, evaluates their vulnerability as vernacular architectural style and therefore, allows to understand and better approach any future intervention on such kinds of structures.* [DOI: 10.5281/zenodo.7492403](https://doi.org/10.5281/zenodo.7492403).
2. **Jiménez Rios, A.** (2016) Simulation of structural behaviour of masonry using discrete element modelling. Master thesis. Polytechnic University of Catalonia, Spain. *The objective of this master thesis is to evaluate the applicability of DEM to simulate the structural behaviour of masonry at a micro and macro-scale level, reproducing the response in terms of deflections, ultimate load and failure modes.* [DOI: 10.5281/zenodo.7492479](https://doi.org/10.5281/zenodo.7492479).
3. **Jiménez Rios, A.** (2014) Comparative analysis of plane frames' structural behaviour considering the effect of end length offsets. Bachelor thesis. University of Guadalajara, Mexico. *The purpose of this thesis was to present a more realistic model to describe the structural response of plane frames by considering the effect that the actual geometrical dimensions of the nodes have in terms of lateral displacements.* [DOI: 10.5281/zenodo.7492520](https://doi.org/10.5281/zenodo.7492520).

Conference posters and presentations:

1. **Jiménez Rios, A.** (2025) Regenerative Structures: Integrating Bamboo, Conservation Engineering, and Digital Twins. Poster presented at the Amazonia Workshop 2025, Belem, Brazil. *Background poster to introduce my research and professional background to other attendees of the Amazonia Workshop 2025.* [DOI: 10.5281/zenodo.15861774](https://doi.org/10.5281/zenodo.15861774).
2. Nela, B., **Jiménez Rios, A.**, Pingaro, M., Reccia, E., & Trovalusci, P. (2022) A limit analysis approach to the assessment of masonry arch structures under the influence of fundamental parameters. Presented at the 11th International Civil Engineering, Infrastructure and Environment Conference (IC-CEIE) 2022, Pristina, Kosovo. *Research on critical parameters affecting the safety levels of piers under the thrust of arches is carried out using an in-house code (ALMA) that uses limit analysis as a rigid block approach to modelling masonry with frictional joints.* [DOI: 10.5281/zenodo.7580693](https://doi.org/10.5281/zenodo.7580693).
3. Nela, B., **Jiménez Rios, A.**, Pingaro, M., Reccia, E., & Trovalusci, P. (2022) Limit analysis for masonry arches: influence of pier texture, arch shallowness, and joint friction. Presented at the XXV CONVEGNO AIMETA – Online Conference. *Using an in-house code that utilizes limit analysis as a rigid block approach to modelling masonry with frictional joints, a study on the crucial parameters impacting the safety level of piers under the thrust of arches is performed.* [DOI: 10.5281/zenodo.7579527](https://doi.org/10.5281/zenodo.7579527).
4. **Jiménez Rios, A.** (2019) Proposal for the sustainable reuse and redevelopment of Newtownbutler's Lanesborough Hotel and former Market House. Poster presented at the "Conservation without Frontiers" Ulster Architectural Heritage & Irish Georgian Society Summer School, Enniskillen, United Kingdom. *The proposal consists in the adaptation of the Market House into a Craft Training and Sustainability Research Centre.* [DOI: 10.5281/zenodo.7496552](https://doi.org/10.5281/zenodo.7496552).

Databases:

1. **Jiménez Rios, A.**, Nogal, M., Plevris, V., Ramirez, R., & L. Petrou, M. (2024) Dataset from the Survey on Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice. Zenodo. *This database contains all the responses from the participants in the survey: Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice.* [DOI: 10.5281/zenodo.12277019](https://doi.org/10.5281/zenodo.12277019).
2. **Jiménez Rios, A.**, L. Petrou, M., Ramirez, R., Plevris, V., & Nogal, M. (2024) Bibliographic Dataset for the Systematic Literature Review on Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice. Zenodo. *This database contains all the bibliographic information found after applying the Search Strategy used for the Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice: Systematic Literature Review.* [DOI: 10.5281/zenodo.10671411](https://doi.org/10.5281/zenodo.10671411).
3. **Jiménez Rios, A.**, Plevris, V., Nogal, M., & Admiraal, W. (2023) Bibliographic Data from the SoTL in Civil and Structural Engineering Systematic Review. Zenodo. *This database contains all the bibliographic*

information found after applying the Search Strategy used for the SoTL in Civil and Structural Engineering Systematic Review. [DOI: 10.5281/zenodo.7951039](https://doi.org/10.5281/zenodo.7951039).

4. Plevris, V., Papazafeiropoulos, G., & **Jiménez Rios, A.** (2023) Dataset of the study: "Chatbots put to the test in math and logic problems: A preliminary comparison and assessment of ChatGPT-3.5, ChatGPT-4, and Google Bard". Zenodo. *This dataset contains the 30 questions that were posed to the chatbots (i) ChatGPT-3.5; (ii) ChatGPT-4; and (iii) Google Bard, in May 2023.* [DOI: 10.5281/zenodo.7940781](https://doi.org/10.5281/zenodo.7940781).
5. **Jiménez Rios, A.**, Ruiz-Capel, S., Plevris, V., & Nogal, M. (2023) Bibliographic Data from the Computational Methods Applied to Earthen Historical Structures Review. Zenodo. *This database contains all the bibliographic information about the 293 records found after applying the Search Strategy used for the Computational Methods Applied to Earthen Historical Structures Review.* [DOI: 10.5281/zenodo.7907457](https://doi.org/10.5281/zenodo.7907457).
6. **Jiménez Rios, A.**, Plevris, V., & Nogal, M. (2023) Bibliographic Data from the Digital Twin Anomaly Detection Decision-Making for Bridge Management Systematic Review. Zenodo. *This database contains all the bibliographic information about the 8673 records found after applying the Search Strategy used for the Digital Twin Anomaly Detection Decision-Making for Bridge Management Systematic Review.* [DOI: 10.5281/zenodo.7548017](https://doi.org/10.5281/zenodo.7548017).
7. **Jiménez Rios, A.**, Pingaro, M., Trovalusci, P., Nela, B., & Reccia, E. (2022) Data from the parametric analysis of masonry buttressed arches with limit analysis subjected to vertical self-weight plus a proportional horizontal live load. Zenodo. *For each one of the simulations performed from the parametric analysis of masonry buttressed arches with limit analysis subjected to vertical self-weight plus a proportional horizontal live load, this database contains a .txt, a .vtk and a .png file. In the .txt file the elapsed time and the collapse multiplier of each simulation can be found. The .vtk file contains all the geometry and displacement values of every masonry buttressed arch. Finally, the .png file presents the collapse mechanism obtained.* [DOI: 10.5281/zenodo.6811858](https://doi.org/10.5281/zenodo.6811858).
8. **Jiménez Rios, A.**, Pingaro, M., Trovalusci, P., Nela, B., & Reccia, E. (2022) Data from the parametric analysis of masonry buttressed arches with limit analysis subjected to vertical self-weight plus a proportional concentrated vertical live load applied at mid-span. Zenodo. *For each one of the simulations performed from the parametric analysis of masonry buttressed arches with limit analysis subjected to vertical self-weight plus a proportional concentrated vertical live load applied at mid-span, this database contains a .txt, a .vtk and a .png file. In the .txt file the elapsed time and the collapse multiplier of each simulation can be found. The .vtk file contains all the geometry and displacement values of every masonry buttressed arch. Finally, the .png file presents the collapse mechanism obtained.* [DOI: 10.5281/zenodo.6811832](https://doi.org/10.5281/zenodo.6811832).
9. **Jiménez Rios, A.**, Pingaro, M., Trovalusci, P., Nela, B., & Reccia, E. (2022) Data from the parametric analysis of masonry buttressed arches with limit analysis subjected to vertical self-weight plus a proportional concentrated vertical live load applied at quarter-span. Zenodo. *For each one of the simulations performed from the parametric analysis of masonry buttressed arches with limit analysis subjected to vertical self-weight plus a proportional concentrated vertical live load applied at quarter-span, this database contains a .txt, a .vtk and a .png file. In the .txt file the elapsed time and the collapse multiplier of each simulation can be found. The .vtk file contains all the geometry and displacement values of every masonry buttressed arch. Finally, the .png file presents the collapse mechanism obtained.* [DOI: 10.5281/zenodo.6811797](https://doi.org/10.5281/zenodo.6811797).
10. **Jiménez Rios, A.**, Pingaro, M., Trovalusci, P., Nela, B., & Reccia, E. (2021) Data from the parametric analysis of masonry pointed arches with limit analysis subjected to vertical self-weight plus a proportional horizontal live load. Zenodo. *For each one of the simulations performed from the parametric analysis of masonry pointed arches with limit analysis, this database contains a .txt, a .vtk and a .png file. In the .txt file the elapsed time and the collapse multiplier of each simulation can be found. The .vtk file contains all the geometry and displacement values of every masonry pointed arch. Finally, the .png file presents the collapse mechanism obtained.* [DOI: 10.5281/zenodo.5146824](https://doi.org/10.5281/zenodo.5146824).
11. **Jiménez Rios, A.**, Pingaro, M., Trovalusci, P., Nela, B., & Reccia, E. (2021) Data from the parametric analysis of masonry pointed arches with limit analysis subjected to vertical self-weight plus a vertical concentrated live load. Zenodo. *For each one of the simulations performed from the parametric analysis of masonry pointed arches with limit analysis, this database contains a .txt, a .vtk and a .png file. In the .txt file the elapsed time and the collapse multiplier of each simulation can be found. The .vtk file contains all the geometry and displacement values of every masonry pointed arch. Finally, the .png file presents the collapse mechanism obtained.* [DOI: 10.5281/zenodo.5146866](https://doi.org/10.5281/zenodo.5146866).

12. **Jiménez Rios, A.**, Pingaro, M., Trovalusci, P., & Reccia, E. (2020) Data from the parametric analysis of masonry panels with limit analysis. Zenodo. *For each one of the simulations performed from the parametric analysis of masonry panels with limit analysis, this database contains a .txt, a .vtk and a .png file. In the .txt file the elapse time and the collapse multiplier of each simulation can be found. The .vtk file contains all the geometry and displacement values of every masonry panel. Finally, the .png file presents the collapse mechanism obtained.* [DOI: 10.5281/zenodo.4320201](https://doi.org/10.5281/zenodo.4320201).
13. **Jiménez Rios, A.** (2019) Cob wallettes' consolidation data. Mendeley Data. *The raw data is divided in three files. The first one contains all the data points collected during the first cycle loading for load (kN), and displacement (mm) for each wallette. The second file contains the same data collected during a second loading cycle (except for wallette W1). Finally, the third one contains data points collected during the consolidation period of five days for load (kN), displacement (mm) and time (min).* [DOI: 10.17632/twbf86wxtx.1](https://doi.org/10.17632/twbf86wxtx.1).
14. **Jiménez Rios, A.** (2019) Cob cylinder's data. Mendeley Data. *The raw data file contains all the data points collected for load (N), test time (s) and displacement (mm) for each cylinder.* [DOI: 10.17632/h8ksd6mvkj.1](https://doi.org/10.17632/h8ksd6mvkj.1).

Open science related and other non-peer reviewed publications:

- Cin Man, A. L., & **Jiménez Rios, A.** (2025) Tourist taxes for Public Realm Conservation in Bath: A Stakeholder Analysis. Zenodo. [DOI: 10.5281/zenodo.17494879](https://doi.org/10.5281/zenodo.17494879).
- Mashal, M., & **Jiménez Rios, A.** (2025) A review of literature and building standards to inform design and construction of earthquake resistant earthen houses in Afghanistan. Preprint. Zenodo. [DOI: 10.5281/zenodo.17056266](https://doi.org/10.5281/zenodo.17056266).
- **Jiménez Rios, A.** & Castilla, F. (2025) Human-Centric Vision, Bridge Design and Engineering. Available from [HERE](#).
- Hawkins, W., Darby, A., & **Jiménez Rios, A.** (2025) Scaling Up Demand for Structural Stone: A Research Perspective, Natural Stone Journal. Available form [HERE](#).
- **Jiménez Rios, A.** et al. (2025) RENEW Manifesto, University of Bath. ISBN 978-1-5262-1030-2. Available form [HERE](#).
- **Jiménez Rios, A.**, Plevris, V., Nogal, M., & Admiraal, W. (2024) SoTL in Civil and Structural Engineering Systematic Review Preprint. Zenodo. [DOI: 10.5281/zenodo.10607158](https://doi.org/10.5281/zenodo.10607158).
- Thongchom, C., Ghamari, A., **Jiménez Rios, A.**, & Karimi, I. (2024) Numerical Analysis and Post-Fire Performance Assessment of a Novel T-Stub Enhanced ConXtech® ConXL™ Moment Connection Preprint. Preprints. [DOI:10.20944/preprints202412.2415.v1](https://doi.org/10.20944/preprints202412.2415.v1).
- **Jiménez Rios, A.**, Nogal, M., Plevris, V., Ramirez, R., & L. Petrou, M. (2024) Towards Enhanced Built Cultural Heritage Conservation Practices: Perceptions on Industry 5.0 Principles and Enabling Technologies Preprint. Zenodo. [DOI: 10.5281/zenodo.12744766](https://doi.org/10.5281/zenodo.12744766).
- **Jiménez Rios, A.**, L. Petrou, M., Ramirez, R., Plevris, V., & Nogal, M. (2024) Industry 5.0 Principles and Enabling Technologies, Towards an Enhanced Conservation Practice: Systematic Literature Review Preprint. Zenodo. [DOI: 10.5281/zenodo.11196879](https://doi.org/10.5281/zenodo.11196879).
- **Jiménez Rios, A.**, L. Petrou, M., Ramirez, R., Plevris, V., & Nogal, M. (2024) Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice: Systematic Literature Review Search Strategy Checklist. Zenodo. [DOI: 10.5281/zenodo.10875793](https://doi.org/10.5281/zenodo.10875793).
- **Jiménez Rios, A.**, L. Petrou, M., Ramirez, R., Plevris, V., & Nogal, M. (2024) Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice: Systematic Literature Review Protocol Checklist. Zenodo. [DOI: 10.5281/zenodo.10877563](https://doi.org/10.5281/zenodo.10877563).
- Arce, A., **Jiménez Rios, A.**, Tomic, I., & Biggs, D. (2024) Structural Analysis in the Sympathetic Restoration and Conservation of the Gopinath Temple, Kathmandu, Nepal. Preprints. [DOI: 10.20944/preprints202403.1578.v1](https://doi.org/10.20944/preprints202403.1578.v1).
- **Jiménez Rios, A.**, Plevris, V., Nogal, M., & Admiraal, W. (2023) SoTL in Civil and Structural Engineering Search Strategy. Zenodo. [DOI: 10.5281/zenodo.7937754](https://doi.org/10.5281/zenodo.7937754).

- **Jiménez Rios, A.,** Plevris, V., Nogal, M., & Admiraal, W. (2023) SoTL in Civil and Structural Engineering Protocol. Zenodo. [DOI: 10.5281/zenodo.7937812](https://doi.org/10.5281/zenodo.7937812).
- **Jiménez Rios, A.** (2023) Learning from the Past: Parametric Analysis of Cob Walls. Preprints. [DOI: 10.20944/preprints202307.0960.v1](https://doi.org/10.20944/preprints202307.0960.v1).
- Plevris, V., Papazafeiropoulos, G., & **Jiménez Rios, A.** (2023) Chatbots put to the test in math and logic problems: A preliminary comparison and assessment of ChatGPT-3.5, ChatGPT-4, and Google Bard. arXiv. [DOI: 10.48550/arXiv.2305.18618](https://doi.org/10.48550/arXiv.2305.18618).
- **Jiménez Rios, A.,** Ruiz-Capel, S., Plevris, V., & Nogal, M. (2023) Computational Methods Applied to Earthen Historical Structures Preprint. Zenodo. [DOI: 10.5281/zenodo.7907825](https://doi.org/10.5281/zenodo.7907825).
- **Jiménez Rios, A.,** Plevris, V., & Nogal, M. (2023) DTADD Systematic Review Preprint. Zenodo. [DOI: 10.5281/zenodo.7673718](https://doi.org/10.5281/zenodo.7673718).
- **Jiménez Rios, A.,** Plevris, V., & Nogal, M. (2023) DTADD Systematic Review Protocol (amended). Zenodo. [DOI: 10.5281/zenodo.7546576](https://doi.org/10.5281/zenodo.7546576).
- **Jiménez Rios, A.,** Plevris, V., & Nogal, M. (2023) DTADD Systematic Review Search Strategy (amended). Zenodo. [DOI: 10.5281/zenodo.7546557](https://doi.org/10.5281/zenodo.7546557).
- **Jiménez Rios, A.,** Pingaro, M., Trovalusci, P., & Reccia, E. (2020). Code from the parametric analysis of masonry panels with limit analysis. Zenodo. [DOI: 10.5281/zenodo.4321939](https://doi.org/10.5281/zenodo.4321939).

Attendance to other academic/research events:

- **Renew Conference** (2026) *The Inaugural Conference on Regenerative Futures: Engineering, Design, Systems and Communities* took place in Bath, UK, and was organized by the Centre for Regenerative Design & Engineering (RENEW). Event website: [HERE](#).
- **BDTIC** (2026) *The 6th Building Digital Twin International Congress* took place in Brussels, Belgium, and was organized by the Building Digital Twin Association (BDTA). Event website: [HERE](#).
- **Bath Digital Festival** (2026) *Bath Digital Festival* took place over 3 days and transformed Bath into a vibrant hub of innovation, creativity and cutting-edge technology. In particular, I attended the “What If Digital Technology Could Open Up Our Heritage?” and “Riding The Next Wave of Digital Innovation” sessions. Event website: [HERE](#).
- **International Seminar on Meteo-Tsunami Research** (2026) *This hybrid event included presentations on meteo-tsunamis in the UK, analysis of the 2025 Kamchatka tsunami, basin geometry effects on tsunami propagation, damage from 2026 storm Ingrid in the UK and meteo-tsunamis as a global scale oceanic hazard. Hosted at the University of Bath.* Event poster: [HERE](#).
- **CCC-ParaSolS Network Event 4** (2026) *This event focused on GPU computing advances as well as data analysis and visualisation training with Paraview. It was hosted at Queen’s University, Belfast, Northern Ireland.* Event website: [HERE](#).
- **Architectural Heritage Symposium** (2026) *This event marked the culmination of the British Academy-funded project: “Ways of Knowing” and “Ways of Doing” for Architectural Heritage Resilience, which focused on: heritage resilience policies; strengthening community engagement; knowledge and skills; and opportunities and impact through tourism.* Project website: [HERE](#).
- **CCC-ParaSolS Network Event 3** (2026) *This event focused on the use of AI in the simulation of particulate solids and was hosted at The University of Manchester.* Event website: [HERE](#).
- **Computing Insight UK** (2025) *CIUK 2025 included two full days of presentations about high performance computing and associated science and research, alongside the CIUK exhibition with fifty exhibitors showcasing the latest hardware and software releases.* Event website: [HERE](#).
- **CoSeC Conference** (2025) *This event reflects the achievements of the past year and the body of research that addresses the increasingly complex scientific computing challenges that we face today.* Event website: [HERE](#).

- **ICCROM Ctrl+S Culture: AI and Heritage in a Digital World International Virtual Conference (2025)** *Ctrl+S Culture was a three-day international online conference exploring how Artificial Intelligence is transforming the ways we preserve, interpret, and engage with cultural heritage.* Event website: [HERE](#).
- **RENEW Symposium (2025)** *The 2025 RENEW Symposium was an afternoon of ideas, case studies and networking hosted by the Centre for Regenerative Design and Engineering.* Event website: [HERE](#).
- **University of Bath Engineering Design and Project Exhibition (2025)** *Annual Engineering Design and Project Exhibition highlights Bath students' inspiring creations.* Event website: [HERE](#).
- **Bath Digital Festival (2025)** *Bath Digital Festival took place over 3 days and transformed Bath into a vibrant hub of innovation, creativity and cutting-edge technology. It was more than just a local event. It was a curated festival designed to give the entire South West region a platform to explore, learn and celebrate advancements in the digital and tech industries.* Event website: [HERE](#).
- **UK Shelter Forum (2025)** *The event's twin themes of 'Building for Climate Change and Extreme Environments', and 'The Ways Forward in Times of Uncertainty' were explored through a series of talks and workshops by Bath researchers and people working with shelters in charities, NGOs and international aid organisations.* Event website: [HERE](#).

10. Academic Experience

2024-Present: Lecturer in Structural Engineering.

Activities: Lecturer at the Department of Architecture & Civil Engineering involved in the following undergraduate and graduate courses:

- MSc Conservation of Historic Buildings.
 - [AR50473 Structural Conservation.](#)
 - [AR50475 Conservation Project.](#)
 - [AR50472 Professional and Research Skills for the Construction Industry.](#)
 - [AR50477 MSc Dissertation.](#)
- BEng and MEng in Civil Engineering/Structural Engineering and Architecture.
 - [AR40301 Conservation Engineering.](#)
 - [AR40443 Advanced Research Dissertation.](#)
 - [AR40392: Natural building materials.](#)
 - [AR40307 Group Design Project - Civil and Architectural.](#)
 - [AR40445 Programming for Design.](#)
 - [AR40443 Advanced Research Dissertation.](#)
 - [AR30315 Dissertations.](#)
 - [AR20526 Engineering Mathematics and Computation 2.](#)
 - [AR10496 Design Skills.](#)
 - [AR10498 – AR10500 Design Project.](#)

Besides, I have participated in construction site visits which are part of the extracurricular activities our students engage with and an important component of the accreditation of our teaching programmes.

I am also a member of the Department's Board of Assessment, where I have reviewed colleagues' exams to ensure high quality standards are maintained.

University: University of Bath, United Kingdom.

2023: Lecturer.

Activities: Co-Lecturer at the Department of Built Environment in the following undergraduate and graduate courses:

- BYTS2500 Concrete Constructions.
- MABY4100 Finite Element Method in Structural Analysis.

University: Oslo Metropolitan University, Norway.

2021: New Master Program on Civil Engineering Creation Committee Member.

Activities: I participated in the committee in charge of designing and implementing a new Master program in the field of Civil Engineering at the Department of Civil and Topography Engineering.

University: University of Guadalajara, Mexico.

- 2019:** Assessment Assistant.
Activities: Support of Trinity Assessment Fellows to map and analyse assessment practices within Schools across the University. Digital badge on programme-focused assessment.
University: Trinity College Dublin, Ireland.
- 2016 – 2019:** Teaching Assistant.
Activities: Teaching assistant at the following engineering courses of the Departments of Civil, Structural and Environmental Engineering and Department of Mechanical Engineering:
- 2E9 Engineering design.
 - 3A2 Structural design.
 - 2E8 Materials.
 - 3E4 Innovation and entrepreneurship.
 - 2E4 Solids and structures.
 - 4E6 Advanced theory of structures.
 - 3B3 Mechanics of solids.
 - C05 Spatial analysis using GIS.
- University:** Trinity College Dublin, Ireland.
- 2018 – 2019:** Workshops.
Activities: Organization and facilitation of three one-day workshops on conservation, vernacular architecture, earthen construction techniques and sustainability.
University: Trinity College Dublin, Ireland.

11. Leadership Roles

- 2025-Present:** [History, Conservation, and Management Cluster](#) Co-Leader.
Department: [Department of Architecture & Civil Engineering](#).
University: [University of Bath](#), United Kingdom.

12. Pedagogical Education

- 2024-2026:** Pathway to Fellowship for Probationary Lecturers.
Description: The Probation Pathway to Fellowship is a work-based curriculum of core activities for lecturers on probation. It is aimed at developing understanding of academic practice within individual, departmental and disciplinary context. Successful completion will lead to HEA Fellowship.
University: University of Bath, United Kingdom.
- 2023:** Teaching and Learning in Higher Education.
Description: Compulsory qualification offer that meets the minimum requirement for basic pedagogical competence for academic staff at universities and colleges:
- Student learning in higher education.
 - Designing teaching in higher education.
 - Assessment and evaluation in higher education.
 - Forms of knowledge in professional education.
 - Research supervision.
- University:** Oslo Metropolitan University, Norway.
- 2022:** 10th Online E-Learning Seminar for University Professors.
Description: 1) Assessment in ICT-supported learning environments, 2) Online game-based learning, 3) The transformation of teaching in online learning environments, 4) Strategies for teaching science and technology online and 5) Strategies for teaching social sciences online.
University: The Catholic University of "Santa Teresa de Jesús de Ávila" (UCAV), Spain.
- 2018:** Reflective Teaching Assistantship Course.
Description: Introduction to current learning theories and teaching practices within higher education. Practical

University: application during an undergraduate lab demonstration.
Trinity College Dublin, Ireland.

2016: CAPSL Postgraduate Teaching Assistant Workshops.
Description:

- How students learn?
- Session planning.

University: Trinity College Dublin, Ireland.

13. Supervising Experience

PhD Theses:

2029: Método de evaluación mecánica para la construcción con tierra mediante pruebas de campo y laboratorio.
Student: Alan Sosa Contreras.
Supervisors: Dr Alberto Muciño Vélez, Dr Luis Fernando Guerrero Baca, Dr Elia Mercedes Alonso Guzmán, and **Dr Alejandro Jiménez Rios.**
Institution: National Autonomous University of Mexico, Mexico.

2028: Conservation of historic fibrous plasters.
Student: Jiahui Yao.
Supervisors: Dr Richard Ball and **Dr Alejandro Jiménez Rios.**
Institution: University of Bath, United Kingdom.

2025: Design of 3D printed steel structures with improved buckling resistance.
Student: Jingbang Pan.
Supervisors: Dr Mark Evernden and **Dr Alejandro Jiménez Rios.**
Institution: University of Bath, United Kingdom.

Master Theses:

2025: How can the ICOMOS-ICCROM Guidance Framework on Post-Conflict Recovery and Reconstruction be revised into a regenerative framework, fostering cultural and social resilience beyond conventional sustainability?
Student: Thaarini Muneswaran.
Supervisor: **Dr Alejandro Jiménez Rios.**
Institution: University of Bath, United Kingdom.

2025: Tourist taxes for Public Realm Conservation in Bath: A Stakeholder Analysis.
Student: Anais Lau.
Supervisor: **Dr Alejandro Jiménez Rios.**
Institution: University of Bath, United Kingdom.

2025: A FOSS-Only HBIM Pipeline: Parametric Modelling of Small Heritage Assets with FreeCAD.
Student: Jing Hao.
Supervisor: **Dr Alejandro Jiménez Rios.**
Institution: University of Bath, United Kingdom.

2023: [Structural damage identification with non-destructive vibrational experiments.](#)
Student: Merhawi Misgina Bayre.
Supervisors: Dr Vagelis Plevris and **Dr Alejandro Jiménez Rios.**
Institution: Oslo Metropolitan University, Norway.

Bachelor Theses:

2024-2025: Finite element analysis model simplification: The roof structure of the Royal Albert Hall.
Student: Emily Munns.
Supervisor: **Dr Alejandro Jiménez Rios.**
Institution: University of Bath, United Kingdom.

Collaborator: [Conisbee](#).

2024-2025: Retrofitting intervention options for the strengthening of masonry arches and barrel vaults.

Student: Robin Wallond.

Supervisor: **Dr Alejandro Jiménez Rios.**

Institution: University of Bath, United Kingdom.

Collaborator: [AtkinsRealis](#).

2024-2025: Repeatable design strategies for the protection of historic buildings in Bath.

Student: Amimah Amod.

Supervisor: **Dr Alejandro Jiménez Rios.**

Institution: University of Bath, United Kingdom.

Collaborator: [Buro Happold](#).

2024-2025: Synthetic data generation for the prototyping of bridge damage detection algorithms.

Student: Matvei Sinden.

Supervisor: **Dr Alejandro Jiménez Rios.**

Institution: University of Bath, United Kingdom.

2024-2025: Impact of different curing conditions on the strength and durability of white cement stabilised rammed earth.

Student: Isabella Marie Newton.

Supervisor: **Dr Alejandro Jiménez Rios.**

Institution: University of Bath, United Kingdom.

2023: Tilting Table Tests of Masonry Assemblies.

Students: Claire Ristow, Onni Heino, Quentin Pierson, Kathy Chhean, and Lara Smit

Context: [European Project Semester \(EPS\)](#).

Supervisor: **Dr Alejandro Jiménez Rios.**

Degree: Bachelor thesis.

Institution: Oslo Metropolitan University, Norway.

14. Examiner Experience

2025: A physics-based probabilistic vulnerability model for the system masonry bridge-soil-river in scour conditions.

Student: Giuseppe Degan Di Dieco.

Advisors: Andre Ramos Barbosa and Raffaele De Risi.

Degree: PhD thesis.

Institution: University of Bristol, United Kingdom.

Role: External Examiner.

2024: Creación de modelos térmicos 3D en edificios y su integración en entornos digitales.

Student: Amanda María Ramón Constantí.

Advisors: Antonio Adán Oliver and Francisco Javier Castilla Pascual.

Degree: PhD thesis.

Institution: University of Castilla La Mancha, Spain.

Role: International External Examiner.

2023: Towards more eco-efficient Ultra High-Performance Concrete.

Students: Berhane Desta Teame and Natnael Russom Afeworki.

Advisors: Mahdi Kioumars and Sarra Drissi.

Degree: Master thesis.

Institution: Oslo Metropolitan University, Norway.

Role: Internal examiner.

2023: Integrating Innovative Technologies for Structural Assessment and Optimisation in Building Design.

Students: Atle Berczelly Schwach and Jacob Olsson Løkhaug.
Advisors: Awais Ahmed and Karoline Baksaas Hersleth.
Degree: Master thesis.
Institution: Oslo Metropolitan University, Norway.
Role: Internal Examiner.

15. Stakeholder Engagement

- 2026:** [Conservation International](#).
Activities: I met Osvaldo Sanchez, Nature-Based Solutions Engineer from Conservation International, at the Department of Architecture and Civil Engineering on the 3rd April. During his visit at University of Bath I presented to him our work at the Department and the different Faculty Research Centres I am part of (RENEW, CAER, The Foundry and CEEDE), shared with him the RENEW Manifesto, and we explored further opportunities for collaboration.
- 2025:** [Institute for Policy Research Policy Fellowship Programme](#).
Activities: Meeting with Fergus Harradence Deputy Director, Construction, Logistics, Airports & Water | Construction, Infrastructure & Energy Directorate at the Department for Business and Trade to discuss how the energy market can deliver the clean energy required by a digitised built environment, what the key trends are for energy generation and storage within the built environment, and what the circular economy looks like in digitised sectors.
- 2024:** [Foster + Partners](#).
Activities: Roger Ridsdill Smith heads-up Foster+Partners' Structural Engineering team and his colleague Rupert Inman were keen to forge stronger links with our Department and visited campus on 28th October to do a show'n'tell about their and our research and explore potential collaborations (UKRI, InnovateUK, PhD, MSc, UG-Dissertations, etc).
- 2024:** [Takenaka Corporation](#).
Activities: Takenaka Corporation's Researcher Dr Jeferson Shigaki - representing the Leads for Digital, Digital Information Management and their Executive Director visited Bath on 02 of September. Whilst they predominantly work in the construction industry, they are leaders in off-site/modular manufacturing & prefabrication. They were looking to exchange experiences and explore potential future collaborations in research, co-development, materials/facilities/data sharing, DfMA/Robotics, etc.

16. Hosting Visitors

- 2025:** In-Person Hosting of Dr Natividad Leonor Garcia Troncoso and Juan Pablo Molina Cedeño.
Affiliation: [Geostructural Sustainable Materials & Innovation Team – GeoSMIT](#), Escuela Superior Politécnica del Litoral (ESPOL), Ecuador.
Activities: During this one-day visit organized in collaboration with the Centre of Excellence in Engineering & Design Education (CEEDE), a mini-symposium and roundtable took place where open discussion took place to identify:
- Common ground in postgraduate education and supervision
 - Student exchange possibilities
 - Joint modules, workshops, and research projects
 - Development of a roadmap for a Bath–ESPOL partnership
- 2025:** In-Person and Online Hosting of Brazilian researchers:
 Dr Ana Lúcia Cerávolo ([UNICEP](#))
 Dr Mônica Martins Andrade Tolentino ([UFVJM](#))
 Dr Ana Regina Mizrahy Cuperschmid ([UNICAMP](#))
 Dr Érico Masiero ([UFSCar](#))
 Professor Márcio Minto Fabricio ([IAU USP](#))
 Dr Juliana Cardoso Esteves ([UNESP](#))
 Dr Beatriz Campos Fialho ([UFSCar](#))
Activities: During this research visit organized by my colleagues Dr Ricardo Codinhoto and Prof Paul

Sheperd, I joined some of the group activities and had one-to-one meetings (Dr Ana Regina Mizrahy Cuperschmid and Dr Beatriz Campos Fialho) to talk about collaboration opportunities on virtual reality, digitalisation, digital twins, BIM, FM, and IoT.

2025: In-Person Hosting of [Dr Francisco Javier Castilla Pascual](#).

Role: Associate Professor.

Affiliation: Department of Applied Mechanics and Project Engineering, University of Castilla – La Mancha, Albacete, Spain.

Activities: During his research visit between May and June at the Department of Architecture & Civil Engineering, University of Bath, Francisco collaborated with me in the REECO-SOIL project, funded by the Centre for Regenerative Design & Engineering for a Net Positive World (RENEW).

17. Mentorship Experience

2025-2026: Academic Advisor.

Students: Marcus Kong, Florence Stewart, Rivaan Nanda, David Burtenshaw, Batool Harb, Joe Dunscombe, Georgie Hartley, Rebecca Dillon, Alvaro Atienza Danobeitia, Iraz Canbay, Emilia Andrews, Grace Ryding, Joseph Moody, Nilufar Montoya Ospina, Kristy Ng, Cameron Oakley, Rhys Lewis, Sutri Ng, Patrick Mason.

Department: Department of Architecture & Civil Engineering

Institution: University of Bath, United Kingdom

2024-2025: Academic Advisor.

Students: Anna Dignam, Cameron Oakley, Grace Ryding, Honor Ash-Dixon, Jack Moore, Joe Dunscombe, Lucy Alexander, Samrat Koirala, Kristy Ng, Sutri Ng, Joe Cooper, Juncheng Yu, Miguel Gil Mejia, Alice Connolly, Billy Rhind-Tutt, Davey Prothero, Romania Long, Thomas Collingridge, Alvaro Atienza Danobeitia, Bobby Leung, Jasmine Long, Lucy Alexander.

Department: Department of Architecture & Civil Engineering

Institution: University of Bath, United Kingdom

2022-Present: MSCA Grant Preparation Mentoring.

Activities: As a successful applicant of the competitive MSCA Individual Fellowship grant scheme, I have provided advice and guidance to people interested in preparing a competitive proposal to get funding from the European Union.

2013-Present: Academic Career Path Mentoring.

Activities: Based on my personal experience, I have provided advice and guidance to people interested in pursuing an academic career in terms of scholarship and grant schemes applications.

18. Academic Administration Experience

2025-Present: Chief Exams Invigilator.

Institution: University of Bath, United Kingdom.

Description: Ensuring that examinations commence, continue, and conclude in an orderly and timely manner, and that students are kept under constant and effective supervision throughout an examination.

2022-2024: Administrator of the Official Website of the Structural Engineering Research Group (SERG) at the Department of Built Environment (DBE), Faculty of Technology, Art, and Design (TDK).

Institution: Oslo Metropolitan University, Norway.

URL: <https://uni.oslomet.no/serg/>

2016-2019: Exams Invigilator.

Institution: Trinity College Dublin, Ireland.

Description: Ensuring that examinations commence, continue, and conclude in an orderly and timely manner, and that students are kept under constant and effective supervision throughout an examination.

19. Editorial Work

2025: Co-Editor Guest.
Journal: Buildings.
Special Issue: [Earth-Based Eco-Efficient Architecture and Construction](#).
Publisher: MDPI.

2025: Handling Editor.
Journal: [Frontiers in Human Dynamics](#).
Publisher: Frontiers.

2025: Co-Editor Guest.
Journal: Buildings.
Special Issue: [Seismic Resilience and Structural Performance of Masonry Structures](#).
Publisher: MDPI.

2024: Leading Guest Editor.
Journal: Infrastructures.
Special Issue: [Advances in Structural Health Monitoring and Industry 5.0 Innovations for Bridge Management and Conservation](#).
Publisher: MDPI.

2023: Leading Guest Editor.
Journal: Frontiers in Built Environment.
Research Topic: [Methods and Applications in Computational Methods in Structural Engineering](#).
Publisher: Frontiers.

2023: Co-Editor Guest.
Journal: Sensors.
Special Issue: [Fault Detection, Diagnosis and Maintenance on Intelligent Transportation System](#).
Publisher: MDPI.

Reviewer in scientific journals:

1. [Structures](#), Elsevier.
2. [Engineering Structures](#), Elsevier.
3. [Construction and Building Materials](#), Elsevier.
4. [Computers and Structures](#), Elsevier.
5. [Automation in Construction](#), Elsevier.
6. [Journal of Building Engineering](#), Elsevier.
7. [International Journal of Critical Infrastructure Protection](#), Elsevier.
8. [Neural Networks](#), Elsevier.
9. [Engineering Applications of Artificial Intelligence](#), Elsevier.
10. [Machine Learning with Applications](#), Elsevier.
11. [Information Fusion](#), Elsevier.
12. [Resilient Cities and Structures](#), Elsevier.
13. [Journal of Open Innovation: Technology, Market, and Complexity](#), Elsevier.
14. [Telematics and Informatics Reports](#), Elsevier.
15. [MethodsX](#), Elsevier.
16. [SoftwareX](#), Elsevier.
17. [Heliyon](#), Elsevier.
18. [Proceedings of the Institution of Civil Engineers - Structures and Buildings](#), ICE.
19. [Proceedings of the Institution of Civil Engineers - Civil Engineering](#), ICE.
20. [Proceedings of the Institution of Civil Engineers - Bridge Engineering](#), ICE.
21. [Proceedings of the Institution of Civil Engineers – Smart Infrastructure and Construction](#), ICE.
22. [Proceedings of the Institution of Civil Engineers - Water Management](#), ICE.
23. [Journal of Structural Design and Construction Practice](#), ASCE.
24. [Journal of Civil Engineering Education](#), ASCE.
25. [npj Heritage Science](#), Springer.
26. [Stochastic Environmental Research and Risk Assessment](#), Springer.

27. [Natural Hazards](#), Springer.
28. [Journal of Infrastructure Preservation and Resilience](#), Springer.
29. [IEEE Transactions on Instrumentation and Measurement](#), IEEE.
30. [IEEE Internet of Things](#), IEEE.
31. [IEEE Access](#), IEEE.
32. [Structural Engineering International](#), Taylor & Francis.
33. [Journal of Asian Architecture and Building Engineering](#), Taylor & Francis.
34. [Applied Artificial Intelligence](#), Taylor & Francis.
35. [International Journal of Digital Earth](#), Taylor & Francis.
36. [Geo-spatial Information Science](#), Taylor & Francis.
37. [Frontiers in Built Environment](#), Frontiers.
38. [Frontiers in Earth Science](#), Frontiers.
39. [Frontiers in Mechanical Engineering](#), Frontiers.
40. [Mathematical Problems in Engineering](#), Wiley.
41. [Advances in Civil Engineering](#), Wiley.
42. [Hormigon y Acero](#), ACHE.
43. [ESTOA Journal of the Faculty of Architecture and Urbanism](#), University of Cuenca.
44. [International Journal of Science and Technology](#), Scientia Iranica.
45. [Heritage](#), MPI.
46. [Smart Cities](#), MDPI.
47. [Architecture](#), MDPI.
48. [Sustainability](#), MDPI.
49. [Applied Sciences](#), MDPI.
50. [Buildings](#), MDPI.
51. [Infrastructures](#), MDPI.
52. [Materials](#), MDPI.
53. [Land](#), MDPI.
54. [Water](#), MDPI.
55. [Oceans](#), MDPI.
56. [Journal of Marine Science and Engineering](#), MDPI.
57. [Processes](#), MDPI.
58. [Symmetry](#), MDPI.
59. [Sensors](#), MDPI.
60. [Drones](#), MDPI.
61. [Information](#), MDPI.
62. [Big Data and Cognitive Computing](#), MDPI.
63. [Multimodal Technologies and Interaction](#), MDPI.
64. [Emergency Management Science and Technology](#), MAP.
65. [Electronic Journal of Structural Engineering](#), EJSEI.

Reviewer for scientific book chapters:

- 2026:** Reviewer.
Book: AI in Education: A Practical Guide for Families, Educators, and Students
Publisher: Springer Nature.
- 2023:** Reviewer.
Book: [Artificial Intelligence and Machine Learning Techniques for Civil Engineering](#).
Publisher: IGI Global.

20. Special Sessions/Mini-Symposia Organizer and Participation in Conference Committees

- 2027:** SS33: Structural Health Monitoring and Industry 5.0 Innovations for Heritage Infrastructure Conservation.
- Conference:** [14th International Conference on Structural Health Monitoring of Intelligent Infrastructures \(SHMII-14\)](#).

- Co-Organizers:** **Alejandro Jiménez Rios** (University of Bath, United Kingdom).
Pier Francesco Giordano (Politecnico di Milano, Italy).
Alberto Barontini (Universià "G d'Annunzio" of Chieti-Pescara, Italy).
- 2026:** Scientific Committee Member and Session Chair.
Conference: [Regenerative Futures Conference: Buildings, Systems and Communities](#)
- 2026:** Scientific Committee Member.
Conference: [XXII International Conference on Building Pathology and Construction Repair \(CINPAR 2026\)](#).
- 2026:** Scientific Committee Member.
Conference: [23rd Seminario Iberoamericano de Arquitectura y Construcción con Tierra \(SIACOT\)](#).
- 2025:** Scientific Reviewing Committee Member.
Conference: [14th International Conference on Structural Analysis of Historical Constructions \(SAHC\)](#).
- 2025:** SS1: Forensic Analysis of Existing Concrete Structures: Advancing NET-ZERO Infrastructure Preservation.
Conference: [2nd International Conference on Net-Zero Built Environment: Innovations in Materials, Structures, and Management Practices \(Net-Zero Future 2025\)](#).
- Co-Organizers:** Afaq Ahmad (Oslo Metropolitan University).
Alejandro Jiménez Rios (University of Bath).
Vagelis Plevris (Qatar University).
- 2025:** SS: Moisture in Multi-Layered Building Envelopes: Mechanisms and Modelling.
Conference: [International Conference on Moisture in Buildings \(ICMB 25\)](#).
Co-Organizers: Rafael Ramirez, (University of Minho, Portugal).
Alejandro Jiménez Rios (University of Bath, United Kingdom).
- 2024:** Scientific Committee Member.
Conference: [ICOMOS Scientific Symposium](#).
- 2024:** SS11: Computational Methods and Digital Twins in the Engineering of Sustainable Buildings, Cities, and Urban Structures.
Conference: [1st International Conference on Net-Zero Built Environment \(Net-Zero Future 2024\)](#).
Co-Organizers: Vagelis Plevris (Qatar University, Qatar).
Awais Ahmed (Oslo Metropolitan University, Norway).
Mohamed El Amine Ben Seghier (Oslo Metropolitan University, Norway).
Alejandro Jiménez Rios (Oslo Metropolitan University, Norway).
- 2024:** MS066: Latest advancements and trends in multi-physics research for civil engineering applications.
Conference: [9th European Congress on Computational Methods in Applied Sciences and Engineering \(ECCOMAS 2024\)](#).
Co-Organizers: Rafael Ramirez, (University of Minho, Portugal).
Alejandro Jiménez Rios (Oslo Metropolitan University, Norway).
Bahman Ghiassi (University of Birmingham, United Kingdom).
- 2024:** MS006: Development and Applications of Computational Methods for Digital Twins.
Conference: [9th European Congress on Computational Methods in Applied Sciences and Engineering \(ECCOMAS 2024\)](#).
Co-Organizers: **Alejandro Jiménez Rios** (Oslo Metropolitan University, Norway).
Vagelis Plevris (Qatar University, Qatar).
Maria Nogal (Delft University of Technology, Netherlands).
- 2024:** SS35: Digital Twins and industry 5.0: towards a sustainable bridge maintenance, safety, and management holistic approach.

- Conference:** [12th International Conference on Bridge Maintenance, Safety and Management \(IABMAS 2024\)](#).
- Co-Organizers:** Alejandro Jiménez Rios (Oslo Metropolitan University, Norway).
Vagelis Plevris (Qatar University, Qatar).
Maria Nogal (Delft University of Technology, Netherlands).
- 2023:** Scientific Committee Member.
- Conference:** [2nd International conference on civil infrastructure and construction \(CIC\)](#).
- 2023:** MS23: Novel data-driven approaches in modelling, assessment, inspection, repairing, strengthening and management of structures.
- Conference:** [9th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering \(COMPDYN\)](#).
- Co-Organizers:** Vagelis Plevris (Qatar University, Qatar)
Alejandro Jiménez Rios (Oslo Metropolitan University, Norway)
German Solorzano (Oslo Metropolitan University, Norway)
Mohamed El Amine Ben Seghier (Institute of Surface Science, Germany).
- 2023:** MS3: Advances in soft computing and optimization methods in engineering.
- Conference:** [15th International Conference on Evolutionary and Deterministic Methods for Design, Optimization and Control \(EUROGEN\)](#).
- Co-Organizers:** Vagelis Plevris (Qatar University, Qatar)
German Solorzano (Oslo Metropolitan University, Norway)
Sadjad Gharehbaghi (Sharif University of Technology, Iran)
Alejandro Jiménez Rios (Oslo Metropolitan University, Norway)

21. *Scientific and professional memberships*

- 2026 – Present:** Academic Member
[The UK Green Building Council](#)
- 2026 – Present:** Cost Action Member
[CA24113 - Global network on Culture Heritage Conservation Under Climate Change \(Go2CHANGE\)](#)
- 2026 – Present:** Member
[CCC-ParaSolS](#).
- 2025 – Present:** Connect Member
[Heritage Network](#).
- 2025 – Present:** Associated member
[Building Digital Twin Association \(BDTA\)](#).
- 2023 – Present:** Member of the:
- [International Scientific Committee on the Analysis and Restoration of Structures of Architectural Heritage \(ISCARSAH\)](#)
 - [International Scientific Committee of Earthen Architectural Heritage \(ISCEAH\)](#).
 - [International Scientific Committee of Heritage Documentation \(CIPA\)](#).
- [International Council on Monuments and Sites \(ICOMOS\)](#).
- 2023 – Present:** Member of the:
- [Technical Committee BEC: Bio-stabilised earth-based construction: performance-approach for better resilience](#).
 - [Technical Committee MAE: Mechanical performance and durability assessment of earthen elements and structures](#).
 - [Technical Committee PEM: Processing of earth-based materials](#).
- [International Union of Laboratories and Experts in Construction Materials, Systems and Structures \(RILEM\)](#).
- 2022 – Present:** Member of the [Task Group 5.8: Resilience of existing structures](#)

- [International Association for Bridge and Structural Engineering \(IABSE\).](#)
- 2019 - Present:** Individual member
- [International Association for Bridge Maintenance and Safety \(IABMAS\).](#)
- 2019 - Present:** General member
- [International Network for Traditional Building, Architecture and Urbanism \(INTBAU\).](#)
- 2019 - Present:** Individual member
- [European Architectural History Network \(EAHN\).](#)
- 2019 - Present:** Member of the Structural Engineering Technical Session
- [Colegio de Ingenieros Civiles del Estado de Jalisco \(CICEJ\).](#)
- 2023 – 2025:** Free member
- [Europa Nostra.](#)
- 2018 - 2020:** Individual member
- [Earth Building UK and Ireland \(EBUKI\).](#)
- 2017 - 2020:** Student member
- [Engineers Ireland \(EI\).](#)

22. *Evaluator of scientific proposals*

- 2025 - Present:** EPSRC Peer Review College Member.
Institution: [UKRI EPSRC.](#)
- 2025 - Present:** Expert Scientific Evaluator Panel Member.
Institution: [Luxembourg National Research Fund.](#)
Proposal type: FNR AFR, Luxembourg.
- 2026:** Expert Scientific Evaluator.
Institution: [University of Guanajuato](#)
Proposal type: Convocatoria Institucional de Investigación Científica 2026, Mexico.
- 2026:** Expert Scientific Evaluator.
Institution: [Research Foundation – Flanders.](#)
Proposal type: FWO 2026 Call for Junior and Senior Postdoctoral Fellowships, Belgium.
- 2025:** Expert Scientific Evaluator.
Institution: [Research Foundation – Flanders.](#)
Proposal type: FWO 2025 Call for Junior and Senior Postdoctoral Fellowships, Belgium.
- 2025:** Expert Scientific Evaluator.
Institution: [University of Lorraine.](#)
Proposal type: LUE IMPACT 2025, France.
- 2025:** Expert Scientific Evaluator.
Institution: [French National Research Agency.](#)
Proposal type: Generic call for proposals – AAPG 2025, France.
- 2025:** Expert Scientific Evaluator.
Institution: [Luxembourg National Research Fund.](#)
Proposal type: FNR CORE, Luxembourg.
- 2025:** Expert Scientific Evaluator.
Institution: [Research Foundation – Flanders.](#)
Proposal type: FWO 2024 Call for Junior and Senior Postdoctoral Fellowships, Belgium.
- 2024:** Expert Scientific Evaluator.
Institution: [Luxembourg National Research Fund.](#)
Proposal type: FNR CORE, Luxembourg.

2024: Expert Scientific Evaluator.
Institution: [Chilean National Agency for Research and Development \(ANID\)](#).
Proposal type: Fondecyt Research Initiation, Chile.

23. *Participation per invitation to scientific seminars/conferences*

2026: From Vernacular to Digital: Structural Engineering and Built Heritage Conservation.
Event: [University of Bath Research Day 2026](#)
Organizers: University of Bath, United Kingdom.

2025: Linux & FOSS: Challenges & Opportunities for Teaching & Research.
Event: [Department of Architecture & Civil Engineering Seminar Series](#).
Organizers: Department of Architecture & Civil Engineering, University of Bath, United Kingdom.

2024: Industry 5.0 and Regenerative Design.
Event: [Smart and Sustainable Infrastructure Team Talk](#).
Organizers: School of Engineering, UWE Bristol, United Kingdom.

2024: Industry 5.0 and the Built Environment.
Event: [IDM Colloquium Series 2024](#).
Organizers: Integral Design and Management Section, Delft University of Technology, Netherlands.

2024: Industry 5.0 Concepts and Enabling Technologies, Towards an Enhanced Conservation Practice.
Event: [ICCROM Fellowship Talks](#).
Organizers: International Centre for the Study of the Preservation and Restoration of Cultural Property (ICCROM), Italy.

2021: Mechanical Properties of Existing Cob Walls.
Event: [NET+WORKING BASEhabitat Hybrid Summer School](#).
Organizers: BaseHabitat, University of Art and Design Linz, Austria.

2021: Conservation of Earthen Vernacular Buildings with an emphasis on minor destructive tests (MDT) and cob mechanical properties.
Event: [Ruins to Riches, Revive, Repurpose, Reimagine Our Traditional Buildings](#).
Organizers: Ulster Architectural Heritage, Armagh City Banbridge & Craigavon and the Department for Communities, United Kingdom.

24. *Other Education*

Summer schools and specialized courses:

2025: Amazonia Workshop.
Description: International workshop on sustainable development of biobased construction materials in Brazil.
Institution: University of Surrey, United Kingdom, and Federal Rural University of the Amazon, Brazil.

2023: 8th CISM-ECCOMAS Advanced School on Scientific Machine Learning in Design Optimization.
Description: The course provided a basic understanding of the classic engineering design process and identified potential usage of artificial intelligence in this context. It covered basic theory of machine learning concepts suitable for (i) geometry representation, (ii) model order reduction, (iii) optimization, and (iv) uncertainty quantification.
Institution: International Centre for Mechanical Sciences (CISM), Italy.

2023: Academic Leadership.

Description: The course is structured into 4 modules: i) Leadership intelligence, ii) Where does your university stand? What is the role of the leader? iii) People-centered leadership, iv) Communication and feedback.

Institution: European Academy of Innovation, Romania.

2023: Structural Health Monitoring for integrity management of civil structures and infrastructures.

Description: The course is structured into 4 modules: i) Fundamentals of SHM, ii) Model-based methods for damage identification, iii) Machine learning methods for condition assessment, iv) SHM for decision support.

Institution: Politecnico di Milano, Italy.

2019: Conservation Without Frontiers International Summer School.

Topic: Survival & Revival: Living Towns and Villages in Cavan and Fermanagh.

Institutions: Ulster Architectural Heritage and Irish Georgian Society, Ireland.

2018: BASEhabitat International Summer School on Earthen and Bamboo Construction.

Topic: Practical workshops in building with earth and bamboo, theory classes on sustainable architecture and networking with people from all around the globe.

University: University of Linz, Austria.

Research oriented:

2025: EPSRC Peer Review Training

Awarded by: Engineering and Physical Sciences Research Council (EPSRC)

Description: Course content: Introduction, assessment principles, application assessment, expert review and the funding service (TFS), EPSRC specific content, additional information.

2024: How to Research and Write Using Generative AI Tools

Awarded by: National Association of State Boards of Accountancy (NASBA)

Description: Top skills covered: Artificial intelligence for business, artificial intelligence for design, writing.

2023: C1 Research and Innovation Project Management: The First Steps.

Awarded by: The Research Council of Norway

Description: Introduction to the life cycle of an EU project; how to start up the project, as well as reporting and management along the way.

2023: A3- Looking for partners and joining consortia

Awarded by: The Research Council of Norway

Description: What type of partners to look for in Horizon Europe? And what characterizes a winning consortium in Horizon Europe?

2023: Research Design.

Awarded by: Elsevier Researcher Academy.

Description: Identifying research gaps, advancing research evaluation strategies.

2022: IOP Peer Review Excellence.

Awarded by: IOP Publishing.

Description: A first-of-its-kind peer review certification programme, Peer Review Excellence: IOP training and certification is co-delivered by top-level physicists and offers a blend of online learning and hands-on peer review experience. Achieving ‘IOP trusted reviewer’ status demonstrates an exceptionally high level of peer review competency.

2022: Book writing.

Awarded by: Researcher Academy, Elsevier.

Description: Why you should publish a book, advice on writing a compelling proposal, ideas for promoting your work.

2022: Technical writing skills.

Awarded by: Researcher Academy, Elsevier.

Description: Practical advice to get you started, guidance on using the system, some golden rules, and common mistakes.

2022: Writing skills.

Awarded by: Researcher Academy, Elsevier.

Description: Information on common writing mistakes, advice on how to avoid those errors, a guide to correct manuscript language.

2022: Fundamentals of manuscript preparation.

Awarded by: Researcher Academy, Elsevier.

Description: An introduction to the publishing process, insights into how to build an article, top tips for writing a great abstract.

2022: Research collaborations.

Awarded by: Researcher Academy, Elsevier.

Description: Advice on how collaborate with industry, guidance on making research collaborations work, an introduction to the importance of collaborations.

2022: Research data management.

Awarded by: Researcher Academy, Elsevier.

Description: Why data sharing is so important, an introduction to data management plans, information about the sharing solutions on offer.

2021: Research funding.

Awarded by: Researcher Academy, Elsevier.

Description: Advice from experts on funding, guidance on writing a great application, an introduction to metrics.

2019: Certified Peer Reviewer Course.

Awarded by: Researcher Academy, Elsevier.

Description: Understanding of the peer review process, advice on writing a helpful peer review report, advice on handling ethical issues concerning peer review.

Completed massive online open courses:

- Grasshopper essential training, LinkedIn, International.
- Rhino 7 essential training, LinkedIn, International.
- Nature-based Solutions for Disaster and Climate Resilience, SDGAcademy, International.
- Getting Started with Mentimeter: Beginner's Course, Mentimeter Academy, Sweden.
- Co-Creation for Policymakers: An Introductory Course, Politecnico di Milano, Italy.
- Preserving Norwegian stave churches, National Technical Norwegian University, Norway.
- The art and science of searching in systematic reviews, National University of Singapore, Singapore.
- Deep learning with TensorFlow, IBM, USA.
- Web applications and command-line tools for data engineering, Duke University, USA.
- Scripting with Python and SQL, Duke University, USA.
- Python and Pandas for data engineering, Duke University, USA.
- Linux and Bash for data engineering, Duke University, USA.
- HTML, CSS, and JavaScript for Web Developers, Johns Hopkins University, USA.
- Machine learning, University of Sandford, USA.
- Construire en terre crue aujourd'hui, Amaco, France.

- Introduction to Python, Microsoft, USA.
- Roman architecture, Yale University, USA.
- The ancient Greeks, Wesleyan University, USA.
- Rome, University of Reading, UK.
- Cultural heritage in transformation, RWTH Aachen University, Germany.
- Arts and heritage management, Bocconi University, Italy.
- Next generation infrastructures, TU Delft, Netherlands.
- The art of structures, Polytechnic University of Lausanne, Switzerland.
- Future smart cities, Swiss Federal Institute of Technology in Zurich, Switzerland.

25. *Professional Experience*

2025 - 2026: Mentoring for Industry Collaboration.

Company: London Structures Lab, United Kingdom.

Activities: The scheme aims to foster mutually beneficial partnerships between academics and industry representatives by:

- Enhancing KE and collaboration.
- Developing strategic partnerships.
- Driving impactful outcomes.
- Expanding engagement and visibility.
- Encouraging reciprocal learning.

01/02/2020 - 31/10/2020: Structural Engineer and leader of R&I.

Company: Xstructuras Ingenieria, Mexico.

Activities: Modelling, analysing, designing, and detailing of steel, concrete, and masonry buildings. Working in and coordinating teams, liaising with architects and clients to ensure the completion and success of projects on time and within budget. Research and implementation of SOA BIM software and working methodologies.

01/09/2014 - 31/08/2015: Structural Engineer.

Company: ALBA Proyecto Estructural, Mexico.

Activities: Modelling, analysing, designing, and detailing of steel, concrete, and masonry buildings. Working in and coordinating teams, liaising with architects and clients to ensure the completion and success of projects on time and within budget.

01/08/2014 - 31/08/2014: Structural Engineer.

Company: Ceromotion, Mexico.

Activities: Modelling, analysing, designing, and detailing of steel, concrete, and masonry buildings.

26. *Professional Certificates*

2015: Certificate on the Analysis and Design of Post-Tensioned Buildings.

Institution: ADAPT Professional Training Institute.

Description: Three-day intensive educational course on the analysis and design of post-tensioned buildings.

27. *Volunteering*

2025-2025: Reviewer.

NGO: [Engineers Without Borders UK](#).

Activities: Reviewed submissions for the EWB-UK Engineering for People Design Challenge.

2013-2014: TECHO Building Volunteer.

NGO: [TECHO](#).

Activities: Built emergency houses for families in need.

28. *Skills*

Leadership:

- Effective communication, task prioritizing, time management, attention to detail, problem-solving, relationship-building, conflict resolution, delegation of tasks and EDI awareness/advocacy.

IT:

- Skills with structural engineering design/analysis software: Rhino, Grasshopper, OpenSees, SAP2000, ETABS, SAFE, ANSYS, ABAQUS, AutoCAD, Revit, Robot, Advance Steel, FormIt, SeismoSignal, Prodisis, Sasid, ALMA, YADE, Paraview.
- Skills with statistical software: Minitab, SPSS.
- Skills with bibliometric software: VOSviewer.
- Skills with programming languages: MATLAB, Python, VBA, XML, HTML, CSS, JavaScript, SQL.
- Skills with developer software: VSCode, Spyder, Vim, Git, GitHub, Docker, Windows Terminal, Anaconda.
- Skills with other software: Word, Excel, PowerPoint, Outlook, Project, Visio, Bookings, Forms, Survey Monkey, Prezi, EndNote, Zotero, LaTeX, Mendeley, JabRef, GIMP, Inkscape, kdenlive, OBS Studio, LibreOffice, Mentimeter.

Languages:

- Spanish: Native.
- English: Advanced.
- French: Intermediate.
- Italian: Beginner.
- Portuguese: Beginner.
- Norwegian: Beginner.